LEMC: Sensor-Free Ego-Motion Compensation

Path-to-Paper Report

Generated 2026-08-30 · PIE · Zhang protocol (τ=16, η=45, TTE 30–60)

Bottom line: Learned LEMC with OBD-magnitude supervision fails the standing-pedestrian variance gate and worsens ADE. **Fixed full-affine ORB compensation** passes the gate (93.1% on PIE, 96.9% on JAAD) and is **ADE-neutral on PIE** (60.5 vs 60.8 px) and **ADE-helpful on JAAD** (62.7 vs 83.1 px, –25%). GT ego speed helps intent (AUC 0.93) but hurts trajectory ADE — consistent with LIM.

1. Executive Summary

Egocentric pedestrian tracks mix pedestrian motion with camera motion. Zhang et al. (TIP 2024) decompose using ego speed as input and soft supervisor; LIM (TITS 2025) deletes speed to avoid causal confusion. This report documents the full path-to-paper run: protocol alignment, sign/variance gate, main experiment grid (3 seeds), ORB extraction, and the pivot from learned residuals to fixed geometric compensation.

Critical finding (Steps 1–2): Attempt-3 LEMC-on (OBD aux, old protocol) — sign gate 0.75% standing flatten; ADE 65.9 vs baseline 43.9. Magnitude–speed correlation (R^2=0.297) is not valid compensation.

2. Protocol & Setup

Setting	Value
Observation τ	16 frames
Prediction η	45 frames
Time-to-event	30–60 frames
Overlap	50%
PIE test windows	1773
Track store	Augmented (median 8 agents/frame)
Seeds	0, 1, 2 (mean ± std reported)

3. Method

Learned LEMC: Multi-agent displacement statistics (ego excluded) → GRU → per-transition residual → cumsum → subtract from ego track. Supervision variants: none, OBD magnitude correlation, learned ORB regression.

Fixed ORB affine (main result): Offline ORB + RANSAC partial affine on consecutive frames with pedestrian boxes masked. Transition target at ego centre (cx,cy) is M@[cx,cy,1] – [cx,cy] — not global translation. Observation and future tracks are compensated into a stabilized frame; GRU predicts; ego-view ADE adds camera shift back.

Translation-only ORB flattened ~21% of standing+moving windows; full affine on the ego point flattened ~89% in a diagnostic (n=300). This is why global (tx,ty) supervision failed.

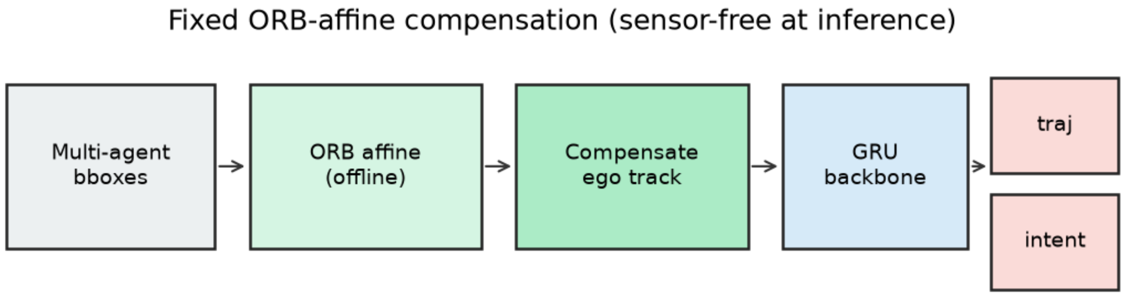

Figure 1. Fixed ORB-affine → GRU pipeline (sensor-free at inference).

4. Main Results (PIE test)

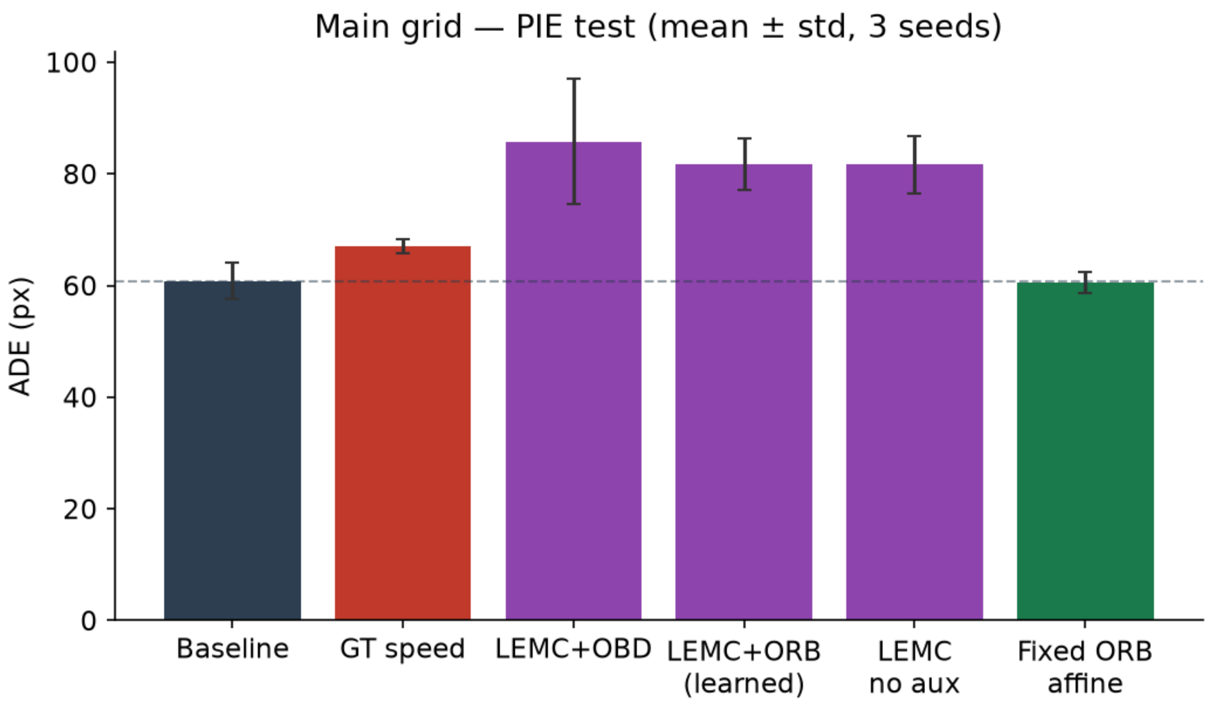

Figure 2. Main grid — ADE (mean ± std, 3 seeds). Fixed ORB matches baseline; learned LEMC variants are worse.

#	Method	ADE	FDE	AUC	F1
1	Baseline	60.81±3.31	121.75±9.87	0.89±0.01	0.74±0.03
2	GT OBD speed	66.96±1.28	136.89±3.02	0.93±0.01	0.85±0.02
3	LEMC + OBD aux	85.77±11.19	164.25±11.33	0.80±0.01	0.60±0.03
4	LEMC + ORB (learned)	81.73±4.61	163.54±7.55	0.81±0.01	0.63±0.00
5	LEMC, no aux	81.68±5.13	159.91±2.39	0.77±0.05	0.64±0.01
6	**Fixed ORB affine**	**60.49±1.83**	**130.28±1.31**	**0.80±0.01**	**0.60±0.01**

Baseline ARB/FRB (seed 0): 35.1 / 61.3 px at 30-frame horizon. Zhang et al. report ADE 17.41 on PIE with I3D vision — gap expected for bbox-only GRU.

5. Sign / Variance Gate

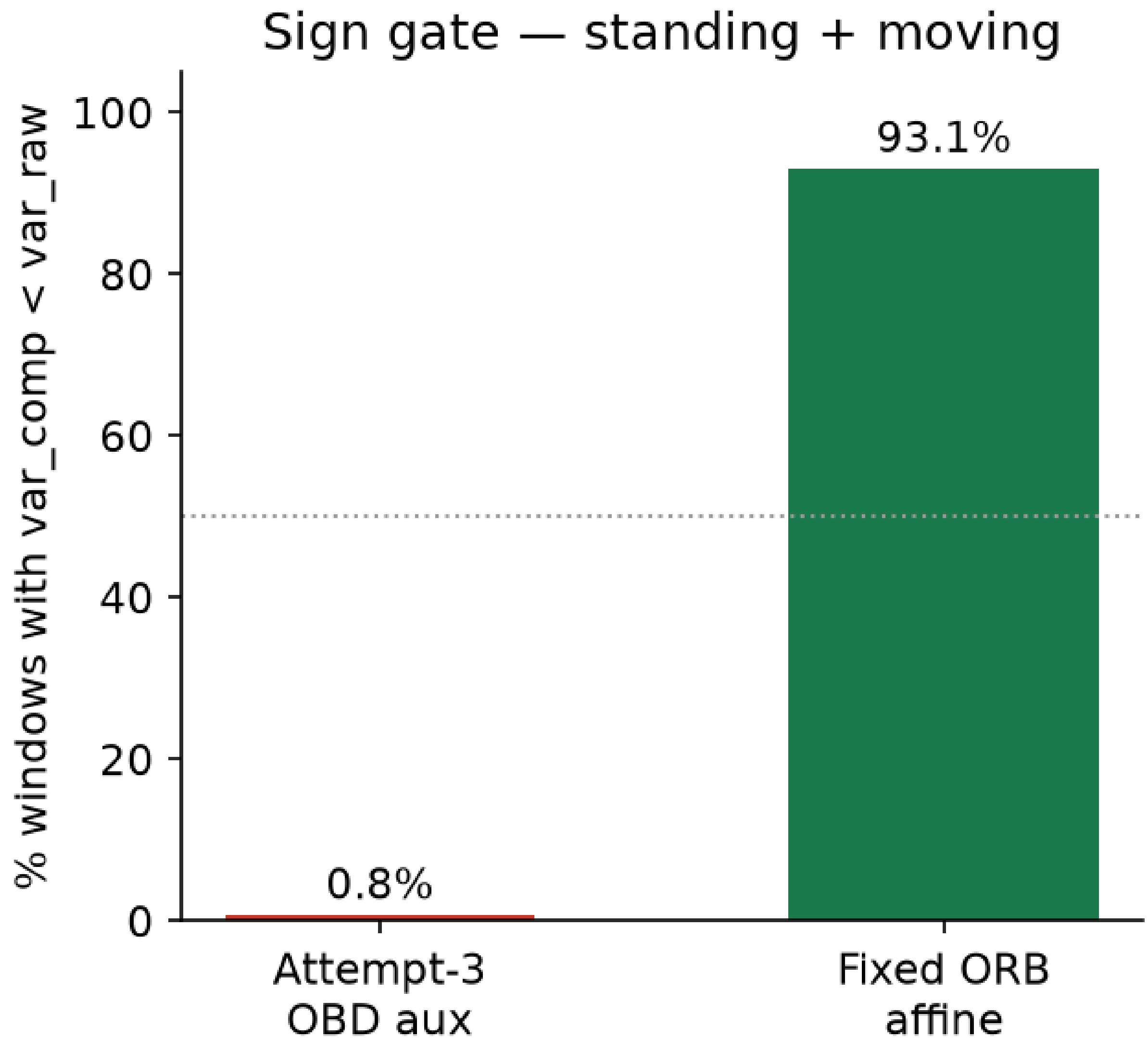

Figure 3. Fraction of windows where compensated track variance < raw variance (standing + moving). Gate: >50% required.

Method	Flatten (all)	Standing + moving
Attempt-3 OBD aux (old proto)	0.7%	0.8% — FAIL
Fixed ORB affine v2	76.7%	93.1% — PASS

6. Qualitative & ORB Validation

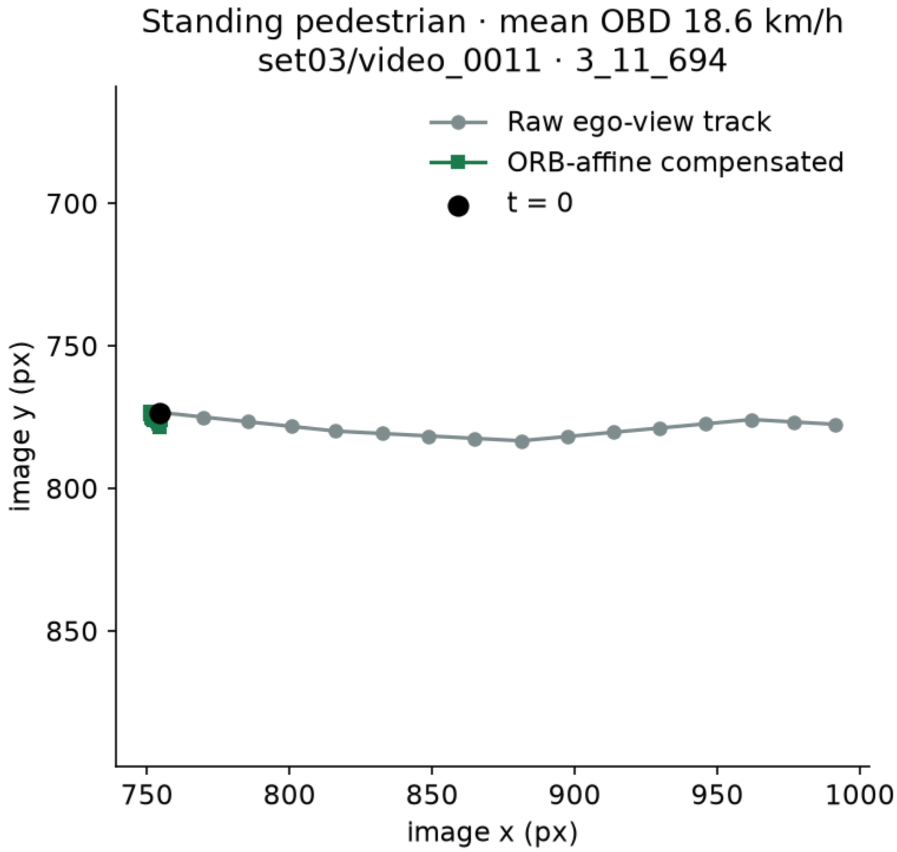

Figure 4. Teaser — standing pedestrian while car moves: raw ego-view track vs ORB-affine compensated.

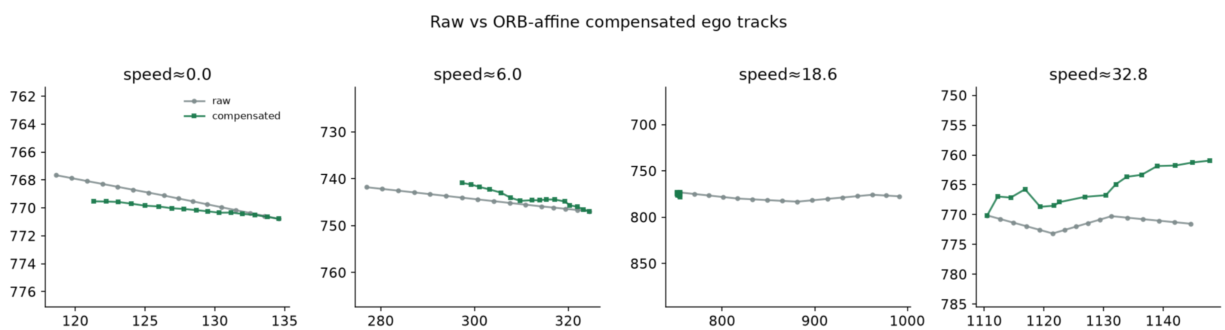

Figure 5. Raw vs compensated tracks across ego-speed percentiles.

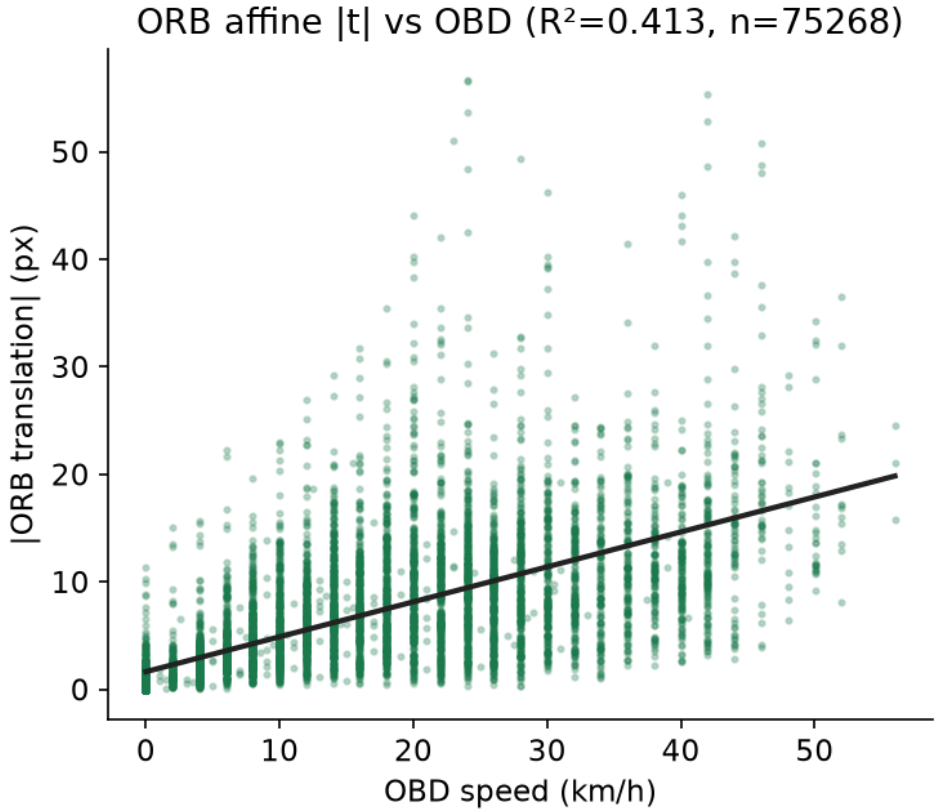

Figure 6. ORB |t| vs OBD speed ($R^2 \approx 0.41$, n=75k transitions). Independent geometric sanity check.

7. Stratified Analysis (seed 0)

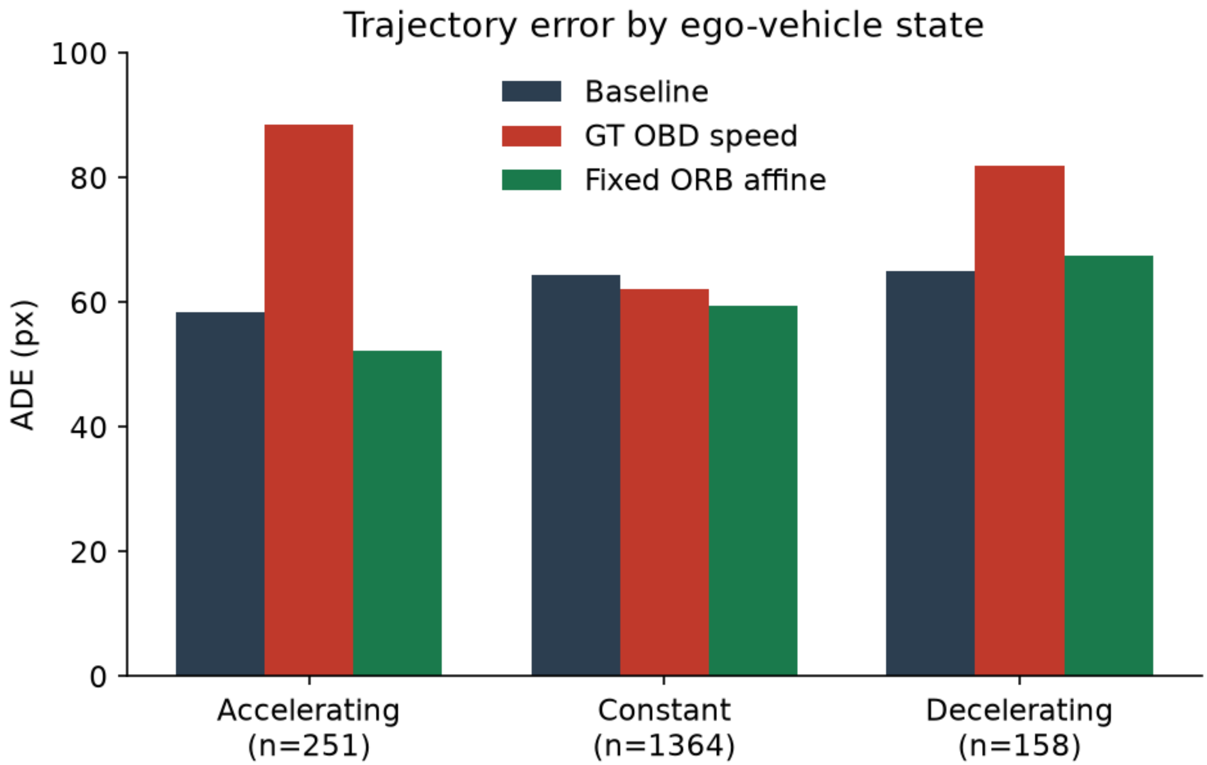

Figure 7. ADE by ego-vehicle state — baseline vs GT speed vs fixed ORB.

Ego state	n	Baseline ADE/FDE	GT speed ADE/FDE	Fixed ORB ADE/FDE
Accelerating	251	58.5 / 127.8	88.6 / 173.2	52.2 / 122.6
Constant	1364	64.5 / 130.6	62.2 / 120.3	59.4 / 130.1
Decelerating	158	65.0 / 139.8	82.0 / 174.7	67.5 / 151.5

Fixed ORB helps most when accelerating (geometry-dominated). GT speed is worst on accelerating/decelerating for ADE while best for intent overall — speed encodes driver reaction more than pedestrian dynamics.

8. Positioning vs Related Work

	Sensor at inference?	Route	JAAD-ready?
Zhang et al. (TIP 2024)	Yes (speed/action)	Two-tower + action-aware loss	Behaviour re-encoding
LIM (TITS 2025)	No (deleted)	Skeleton + adversarial scrub	Yes
Fixed ORB (ours)	**No**	**Offline ORB affine → compensate**	**Yes**
Learned LEMC	No	Multi-agent residual	Failed gate here

9. JAAD Results (sensor-free, no OBD)

JAAD 2.0: 323 videos, 336/52/295 train/val/test windows (TTE 30–60, 50% overlap). No OBD speed — sensor-free only. ORB affines extracted from 124 scenes with images available. 3 seeds run on CPU (training ~10s/seed; small dataset).

Method	ADE	FDE	ARB	FRB	AUC	F1	Sign gate (s+m
Baseline GRU	83.10±2.00	156.25±4.31	44.76±2.15	71.23±2.08	0.47±0.05	0.90±0.00	—
Fixed ORB affine	**62.68±2.34**	**120.11±2.82**	**34.80±2.25**	**54.17±0.91**	**0.44±0.00**	**0.90±0.00**	**96.9%**

JAAD ADE drops **25%** (83.1 → 62.7 px). Unlike PIE (ADE-neutral), JAAD benefits from affine compensation — consistent with stronger and more variable camera motion across JAAD's diverse clip conditions. Sign gate 96.9% standing+moving (all windows 71.9%).

10. Limitations & Next Steps

Single GRU backbone; no on-vehicle latency study beyond CPU note (1.58 ms/seq); learned LEMC does not match ORB under joint training; ORB depth confound; hold-last for missing future affines.

Next: wire \cite{} calls into paper/main.tex; compile blind PDF; outsider read; submit to IEEE IV 2026.

11. Reproducibility

python scripts/09b_extract_orb_affine.py → python scripts/12_fixed_orb_affine.py --seed 0 → bash scripts/10_run_grid.sh → python scripts/13_make_figures.py → python scripts/generate_paper_report.py